

ETF Cases and a Rational Blow-Up Reduction for Basso's Orthogonal Projection Question

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June 19, 2026

Status

This packet records a likely valid partial result for Question 1.1 in Giuliano Basso, *Almost minimal orthogonal projections*, arXiv:2001.08698. It does not settle the question for all dimensions. It proves the question for every dimension admitting a real maximal equiangular tight frame, gives the concrete known cases $n = 1, 2, 3, 7, 23$, and isolates a rationality criterion that would turn Basso's almost-positive construction into an exact construction.

Source Question

Basso asks the following question in the introduction.

Regular triangles and twelve regular pentagons as its faces, has the property that $\Pi(Y_3, \ell_1^6) = \Pi(E_{dod}) = \phi$. It is not hard to check that the orthogonal projection $P: \ell_1^6 \rightarrow Y_3$ is minimal, so $\|P\| = \Pi(Y_3, \ell_1^6) = \phi$. Thus, if $\Pi_3 = \phi$, then $\|P\| = \Pi(Y_3, \ell_1^6) = \Pi_3$. In view of these examples, the following question arises naturally:

Question 1.1. *Fix an integer $n \geq 1$. Does there exist an n -dimensional Banach space $E \subset \ell_1^d$ for some integer $d \geq n$ such that the orthogonal projection $P: \ell_1^d \rightarrow E$ is minimal and $\|P\| = \Pi(E, \ell_1^d) = \Pi_n$?*

We suspect that Question 1.1 has a positive answer for $n = 3$. However, for $n \geq 4$ the general picture is unclear. It appears reasonable to expect that Question 1.1 has only a positive answer for certain integers n .

1.2 Main result

Our main result implies that Question 1.1 has an ‘almost’ positive answer for every integer $n \geq 1$.

Here $\Pi(E, F)$ denotes the relative projection constant, Π_n denotes the maximal real projection constant among n -dimensional Banach spaces, and the orthogonal projection is taken with respect to the Euclidean structure on $\mathbb{R}^d$.

Background from Basso

For a symmetric sign matrix S of size N with diagonal entries equal to 1, let

$$\pi_n(S) = \lambda_1(S) + \cdots + \lambda_n(S),$$

where the eigenvalues are ordered decreasingly. Basso proves

$$\sup_{N \geq n} \sup_{S \in \mathcal{S}_N} \frac{1}{N} \pi_n(S) = \Pi_n.$$

He also proves that if this supremum is attained by some finite sign matrix $S \in \mathcal{S}_N$, then Question 1.1 has a positive answer in dimension n [1, Theorem 1.3 and Proposition 1.4].

Idea of the Proof

There are two ways to make Basso's unweighted sign-matrix supremum attain its value.

First, a real maximal equiangular tight frame produces a rank- n orthogonal projection P whose entries have one diagonal magnitude and one off-diagonal magnitude. Hence P is an affine function of its sign matrix S . The top n eigenvalues of S can then be computed exactly, and the 2022 theorem of Deregowska and Lewandowska says that this value is the maximal projection constant Π_n whenever a real maximal ETF exists.

Second, Basso's proof already gives an exact weighted maximizer $\sqrt{D}S\sqrt{D}$. If the diagonal weights of D are rational, replacing the i -th vertex by a block of size proportional to D_{ii} produces an unweighted sign matrix whose nonzero spectrum is the spectrum of $N\sqrt{D}S\sqrt{D}$. Thus the unweighted supremum is attained exactly.

ETF Subcases

Theorem 1 (Maximal ETF criterion). *Suppose there is a real maximal equiangular tight frame in $\mathbb{R}^n$, that is, an ETF with*

$$N = \frac{n(n+1)}{2}$$

vectors in $\mathbb{R}^n$. Then Basso's Question 1.1 has a positive answer for this integer n .

Proof. Let $x_1, \dots, x_N$ be unit vectors forming a real ETF in $\mathbb{R}^n$, with $N = n(n+1)/2$. Put

$$U = \sqrt{n/N} [x_1 \cdots x_N],$$

so $UU^t = I_n$, and let $P = U^t U$. Then P is an $N \times N$ orthogonal projection of rank n . Set

$$a = \frac{n}{N}, \quad \alpha = \sqrt{\frac{N-n}{n(N-1)}}, \quad b = a\alpha.$$

The ETF property gives

$$p_{ii} = a, \quad |p_{ij}| = b \quad (i \neq j).$$

Let $S \in \mathcal{S}_N$ be the sign matrix of P , with $s_{ii} = 1$ and $s_{ij} = \text{sign}(p_{ij})$ for $i \neq j$. Then

$$P = bS + (a-b)I_N.$$

Since P has eigenvalue 1 with multiplicity n and eigenvalue 0 with multiplicity $N - n$, the matrix S has eigenvalue

$$\frac{1 - a + b}{b}$$

with multiplicity n , and eigenvalue

$$-\frac{a - b}{b}$$

with multiplicity $N - n$. Therefore

$$\frac{1}{N}\pi_n(S) = \frac{n}{N} \frac{1 - a + b}{b} = \frac{n}{N} + \frac{N - n}{N\alpha} = \frac{n}{N} \left(1 + \sqrt{\frac{(N - 1)(N - n)}{n}} \right).$$

For $N = n(n + 1)/2$ this is

$$\frac{2}{n + 1} \left(1 + \frac{n - 1}{2} \sqrt{n + 2} \right).$$

Deregowska and Lewandowska prove that the real maximal projection constant $\lambda_{\mathbb{R}}(n)$ equals this number whenever a real maximal ETF exists [2, Theorem 2.5]. This is Basso's Π_n in the real notation. Hence S attains Basso's supremum

$$\sup_{M \geq n} \sup_{T \in \mathcal{S}_M} \frac{1}{M} \pi_n(T) = \Pi_n.$$

Basso's Proposition 1.4 now supplies an n -dimensional subspace $E \subset \ell_1^N$ whose orthogonal projection is minimal and satisfies

$$\|P_E\| = \Pi(E, \ell_1^N) = \Pi_n.$$

This is precisely a positive answer to Question 1.1 for this n . □

Corollary 1 (Concrete dimensions). *Basso's Question 1.1 has a positive answer for*

$$n = 1, 2, 3, 7, 23.$$

Proof. The case $n = 1$ is immediate from the identity projection on ℓ_1^1 . The real maximal ETF cases listed by Deregowska and Lewandowska are $n = 2, 3, 7, 23$ [2, after Theorem 2.5]. Applying the preceding theorem proves the claim. □

Rational-Weight Reduction

Theorem 2 (Exact rational blow-up criterion). *Fix $n \geq 1$. Suppose there are $m \geq n$, a sign matrix $S \in \mathcal{S}_m$, and a diagonal matrix*

$$D = \text{diag}(d_1, \dots, d_m), \quad d_i > 0, \quad \sum_i d_i = 1,$$

such that every d_i is rational and

$$\pi_n(\sqrt{D} S \sqrt{D}) = \Pi_n.$$

Then Basso's Question 1.1 has a positive answer for this n .

Proof. Choose positive integers $k_1, \dots, k_m$ and let $N = \sum_i k_i$ so that $d_i = k_i/N$. Let $\tilde{S} \in \mathcal{S}_N$ be the block blow-up of S in which vertex i is replaced by a block of size k_i , and the (i, j) -block is constantly equal to s_{ij} .

Let $H \subset \mathbb{R}^N$ be the subspace of vectors that are constant on each block, and write

$$u_i = \frac{\mathbf{1}_{\text{block } i}}{\sqrt{k_i}} \quad (1 \leq i \leq m).$$

The vectors u_i form an orthonormal basis of H . The subspace H is invariant under $\tilde{S}$, and the matrix of $\tilde{S}|_H$ in this basis is

$$(s_{ij} \sqrt{k_i k_j})_{i,j} = N \sqrt{D} S \sqrt{D}.$$

The orthogonal complement $H^\perp$ consists of vectors whose coordinates sum to zero on each block, and $\tilde{S}$ annihilates $H^\perp$. Thus the spectrum of $\tilde{S}$ is the spectrum of $N \sqrt{D} S \sqrt{D}$ together with additional zeros.

Consequently

$$\frac{1}{N} \pi_n(\tilde{S}) \geq \pi_n(\sqrt{D} S \sqrt{D}) = \Pi_n.$$

Basso's formula for the unweighted supremum gives the reverse inequality for every sign matrix, so equality holds:

$$\frac{1}{N} \pi_n(\tilde{S}) = \Pi_n.$$

Therefore $\tilde{S}$ attains Basso's supremum. Proposition 1.4 of [1] again gives the required subspace of ℓ_1^N and a minimal orthogonal projection of norm Π_n . $\square$

Verification Notes

No numerical computation is used. The ETF calculation only uses the standard Gram-matrix form of a tight ETF and the exact equality statement for maximal projection constants from Deregowska–Lewandowska. The rational blow-up argument is the exact version of the approximation-by-rational-weights step in Basso's proof of the almost-positive theorem.

The key point in both arguments is not merely that large projection constants are attained, but that Basso's unweighted sign-matrix supremum is attained. That is exactly the hypothesis needed to invoke Basso's Proposition 1.4.

Bounded novelty checks were performed against the run indexes, the parsed local source of arXiv:2001.08698, exact title/question web searches, and searches for later projection-constant and maximal-ETF papers. No full all- n resolution of Question 1.1 surfaced.

Limitations

This packet is not a full solution. It proves Question 1.1 for the maximal ETF dimensions and gives an exact rationality criterion for the general weighted maximizer. Basso's Lemma 3.7 guarantees a positive diagonal maximizer, but not rational diagonal entries. Without rational weights the block blow-up gives only the approximation already exploited in the source paper.

Human Review Recommendation

Review two points. First, check the affine relation $P = bS + (a - b)I_N$ for the normalized ETF Gram projection and the resulting eigenvalue computation. Second, check that the block blow-up has kernel $H^\perp$ and compression $N\sqrt{D}S\sqrt{D}$ on block-constant vectors. If both checks pass, the reductions to Basso's Proposition 1.4 are direct.

References

- [1] G. Basso, *Almost minimal orthogonal projections*, arXiv:2001.08698, 2020.
- [2] B. Deregowska and B. Lewandowska, *A simple proof of the Grunbaum conjecture*, arXiv:2206.09454, 2022.